A Lieb–Robinson bound for the Kogut–Susskind Hamiltonian, with velocity $\propto Nc/g^2$

The Hamiltonian of the obligations map has exactly the structure for which a Lieb–Robinson bound survives an unbounded generator: its unbounded part is a sum of commuting single-link terms, and its interaction is a bounded finite-range sum of plaquette terms. Passing to the interaction picture generated by the electric term leaves a time-dependent bounded interaction of unchanged norm and range, so the standard theorem applies and gives, for observables A on a set X of links and B on Y,

$$\big\|[A(t),B]\big\| \;\le\; 2\|A\|\,\|B\|\,|X|\,e^{-\left(d(X,Y)-v_{\mathrm{LR}}|t|\right)/a}, \qquad v_{\mathrm{LR}} \;\le\; C_{\mathrm{LR}}aJ/\hbar \;\le\; 16C_{\mathrm{LR}}Nc/g^2,$$

where $J = \max_\ell \sum_{p\ni\ell}\|w_p\| \le 16N\hbar c/(g^2a)$ on the periodic three-dimensional spatial cubic lattice. The finite constant C_{LR} depends on the graph metric and theorem norm, including four-link support factors; it is independent of volume and coupling. The earlier $C_{\mathrm{LR}} = 2e$ was not derived from the metadata-only source citations and is not certified here.

With a separately established gap and vacuum hypotheses the standard clustering theorem applies. Its bound retains the microscopic range; $\hbar v_{\mathrm{LR}}/\Delta$ alone is not a general correlation-length bound. A divergent velocity upper estimate proves no superluminality. Locality is also not by itself a gap-stability theorem.

1. The structure that makes the unbounded term harmless

Take the links as the sites of the lattice and write, as in the obligations map §1,

$$H = H_{\mathrm{E}} + W, \qquad H_{\mathrm{E}} = \frac{\hbar c\, g^2}{2a}\sum_{\ell\in\mathcal{E}}(-\Delta_\ell), \qquad W = \sum_{p\in\mathcal{P}} w_p, \quad w_p = \frac{2\hbar c}{a\, g^2}(N - \operatorname{Re}\operatorname{tr} U_p).$$

Three properties are all that the argument uses.

1. *The unbounded part is single-site and commuting.* $-\Delta_\ell$ acts on the single factor $L^2(G)$ of link ℓ, and the terms for different links commute. Each generates a one-parameter unitary group $e^{-is c g^2 \Delta_\ell/(2a)}$ whose adjoint action preserves the algebra of that link.
2. *The interaction is bounded.* $0 \le w_p \le \frac{4N\hbar c}{ag^2}$ in operator norm for $SU(N)$, since $|\operatorname{Re}\operatorname{tr} U| \le N$; for a general compact G in a faithful representation of dimension $\dim_{\mathrm{f}}$ replace N by $\dim_{\mathrm{f}}$.
3. *The range is one plaquette.* Each w_p acts on the four links of p; a given link lies in exactly four plaquettes in three spatial dimensions (two planes containing its direction, two plaquettes on either side in each). Hence

$$J := \max_\ell \sum_{p\ni\ell}\|w_p\| \;\le\; 4\cdot\frac{4N\hbar c}{ag^2} = \frac{16N\hbar c}{a\,g^2}.$$

2. The reduction and the bound

Proposition 1. Let $\tau_t(A) = e^{itH/\hbar}Ae^{-itH/\hbar}$. For A supported on a set X of links and B on a disjoint set Y, with $d(X,Y)$ the lattice distance,

$$\big\|[\tau_t(A),B]\big\| \;\le\; 2\|A\|\,\|B\|\,|X|\,\exp\Big[\frac{v_{\mathrm{LR}}|t| - d(X,Y)}{a}\Big], \qquad v_{\mathrm{LR}} \le C_{\mathrm{LR}}aJ/\hbar.$$

Proof. Let $U_{\mathrm{E}}(t) = e^{-itH_{\mathrm{E}}/\hbar}$ and define the interaction-picture interaction $\tilde w_p(t) = U_{\mathrm{E}}(t)^* w_p U_{\mathrm{E}}(t)$. Since H_{E} is a sum of commuting single-link terms, $U_{\mathrm{E}}(t)$ factorizes over links and its adjoint action preserves each link algebra, so $\tilde w_p(t)$ is supported on the same four

links and $\|\tilde{w}_p(t)\| = \|w_p\|$ for all t. The full evolution factorizes as $e^{-itH/\hbar} = U_{\mathrm{E}}(t)\,\tilde{U}(t)$ with $\tilde{U}$ generated by the time-dependent bounded interaction $\tilde{W}(t) = \sum_p \tilde{w}_p(t)$, and conjugation by $U_{\mathrm{E}}(t)$ preserves supports and norms. The Lieb–Robinson theorem for time-dependent bounded finite-range interactions (Lieb–Robinson, Commun. Math. Phys. 28 (1972) 251; Nachtergaele–Sims, Commun. Math. Phys. 265 (2006) 119, in the form used for the clustering theorem; metadata level) applies to $\tilde{W}$ with the constant J of Section 1, and transferring back through U_{E} gives the stated volume-uniform scaling. A numerical prefactor requires fixing the theorem norm and graph-distance convention, not only J. □

The same argument is the one used for oscillator lattices, where the unbounded on-site harmonic part is treated exactly and the bounded anharmonic interaction carries the velocity (Nachtergaele–Raz–Schlein–Sims, Commun. Math. Phys. 286 (2008) 1073; metadata level). The gauge constraint plays no role: the bound holds on the full Hilbert space and restricts to the physical subspace, which is invariant under H.

Remark (what fails without the structure). If the electric term coupled two links, its unboundedness would enter the iteration and no bound of this form would follow. The single-link form of $\sum_a E^a_\ell E^a_\ell$ is therefore load-bearing, and it is the same feature that made the strong-coupling hypothesis check of the T2 note work.

3. Clustering and the microscopic range

The strong-coupling theorem, with its infinite-volume construction and vacuum hypotheses, already gives exponential clustering. The standard gapped-Hamiltonian argument has a conservative range-dependent form

$$\xi \le C\left(a + \frac{\hbar v_{\mathrm{LR}}}{\Delta}\right).$$

The constant depends on the locality norm and metric. The range term cannot be discarded uniformly when the quotient tends to zero. With $\Delta \ge \gamma C_2 g^2 \hbar c/(2a)$, the algebraic consequence is

$$\frac{\hbar v_{\mathrm{LR}}}{\Delta} \le \frac{32 C_{\mathrm{LR}} a N}{\gamma C_2 g^4}.$$

This controls the dynamical quotient, not ξ by itself. The earlier claim $\xi = O(ag^{-4})$ is withdrawn. A plaquette-tube weight of order $g^{-4d/a}$ instead has inverse decay length of order $\log g/a$. Nachtergaele–Sims, *CMP* **265** (2006) 119, and Hastings–Koma, *CMP* **265** (2006) 781, are retained at metadata level; their theorems have their own hypotheses and numerical constants.

4. Weak coupling and physical causality

The locality upper estimate becomes weak as $g \to 0$. Comparing that coefficient with c determines no actual propagation speed. Continuum relativistic locality requires a reconstructed physical theory.

Even the formal box-crossing comparison distinguishes a bound from an attained speed. Substituting an order Nc/g^2 coefficient and the conditional zero-mode energy $g^{2/3}\hbar c/L$ compares $Lg^2/(Nc)$ with $L/(cg^{2/3})$. The first formal time is smaller when $g^{8/3} < N$, up to constants. This proves neither causal connection of the box on that time scale nor a nonzero-mode truncation theorem.

5. Gap stability is a separate input

Locality is one ingredient of several stability theorems. Their further ground-space and perturbation-norm conditions must be checked; a frustration-free/topological-order theorem

cannot be applied merely because a locality bound is available. Finite-coupling Kogut–Susskind electric and magnetic terms have no common minimizing state. The strong-coupling stability argument uses its own product reference and bounded perturbation.

No general openness theorem for the whole gapped coupling set follows here. Naming frustration-freeness does not exhaust the missing conditions.

6. Consequence for STATE

Treating commuting unbounded single-link terms exactly leaves bounded plaquette interactions with unchanged supports and norms. This supplies volume-uniform locality of order Nc/g^2. Numerical prefactors, physical propagation, clustering lengths, stability and continuum limits retain separate hypotheses.